

Simultaneous choice: review before calibration

6 September 2026 | Experimental specification | Full overnight search stopped

The lifecycle calculation is reproducible. A successful recalibration has not yet been established. Two full histories reproduce all twelve target rows, all parameters, 253 historical entries and seventeen standard graphs exactly. The checkpoint and consumption-reporting repairs preserve the economic choices and fit.

The starting point fits poorly. Its objective is 485.459. The ownership gap between parents and nonparents is 0.487 percentage points, against 16.766 in the data. The first-birth housing response is 0.334 rooms, against 0.720. These are values at starting parameters, not the result of a calibration search.

The economic issue to discuss is the nesting restriction. The experiment groups waiting and attempting a birth within each tenure. The outer shock scale is κ ; the inner scale is $\lambda\kappa$, with $0 < \lambda \leq 1$. For example, $\kappa = 2$ and $\lambda = 0.8$ imply an inner scale of 1.6. This links tenure and fertility dispersion. Both must be re-estimated jointly with the other nine parameters.

Simultaneous tastes still require a conception contract. The household chooses tenure and whether to attempt a birth together, after observing the joint tastes. Whether conception succeeds is learned afterward. Housing size, consumption and saving may adjust within the chosen tenure. The same nesting parameter applies across birth orders; this is an experimental restriction.

Policy-stage verification: stopped at supply expansion, 2027. The baseline passes both dates. Supply passes 2023, then fails an occupied-value check in 2027. LTV and tax have not run. The exhaustive-saving diagnosis is on page 5.

Proposed next step after numerical verification and our specification discussion: search all eleven parameters against the unchanged twelve targets and weights; reserve time for a local sensitivity matrix, two exact final repetitions and full policy paths. Having twelve moments for eleven parameters is not an identification test. Retain the current expectation method for this run and discuss perfect foresight separately. No production replacement has been made.

Recommendation: use exhaustive saving in the isolated experiment and repeat the complete history/policy smoke before launching the calibration. The fixed-price diagnosis establishes a promising repair, not a verified equilibrium path.

Proposed resource limit: twelve cluster workers, at most 360 historical cases including final checks, up to nine hours searching within twelve hours total. The current optimizer takes 12-14 minutes per history, implying roughly 6-7 hours for 360 cases with twelve workers. Exhaustive saving must be timed on a complete history; the case count may need to fall to preserve final checks and the morning cutoff.

Complete target fit at the starting point

Every original target is retained. Shares and share gaps are fractions: 0.01 equals one percentage point. The objective is the sum of weight $\times$ gap²; the weights do not all represent estimated sampling precision.

Moment	Target	Model	Gap	Weight	Loss
Completed fertility	1.918	1.90219	-0.0158118	1425.74	0.356455
Childlessness	0.188	0.214244	0.0262438	17180.7	11.833
Mean age at first birth	26.0446	25.5939	-0.450769	44.4444	9.03078
Share first births age 30+	0.260327	0.193331	-0.066996	10000	44.8846
Rooms response to first birth	0.720246	0.333599	-0.386647	137.565	20.5655
Rooms: 3+ minus 1-2 children	0.3677	0.326202	-0.0414979	2958.51	5.09479
Parent/nonparent ownership gap	0.167662	0.00487474	-0.162787	14229.6	377.078
Prime-age ownership	0.575472	0.498535	-0.0769374	1207.85	7.14969
Mean occupied rooms	5.77997	5.61864	-0.161331	11.9732	0.311635
Wealth / annual earnings	6.8731	6.96277	0.0896677	6.28767	0.0505547
Bequest flow / wealth	0.0088	0.00805396	-0.000746041	5.16529e+06	2.87488
Old-age wealth p90 / p50	3.44811	3.11741	-0.330697	56.9598	6.22915

Total objective: 485.459247036. The starting ownership-gap miss contributes 377.078, or 77.7% of the total. A small fit improvement in the verification probes is not a searched optimum.

Parameters, search bounds and fixed restrictions

Starting values of the eleven free parameters. The two shock parameters are experimental starting choices; the remaining starting values come from the inherited candidate. None is presented as a newly estimated optimum.

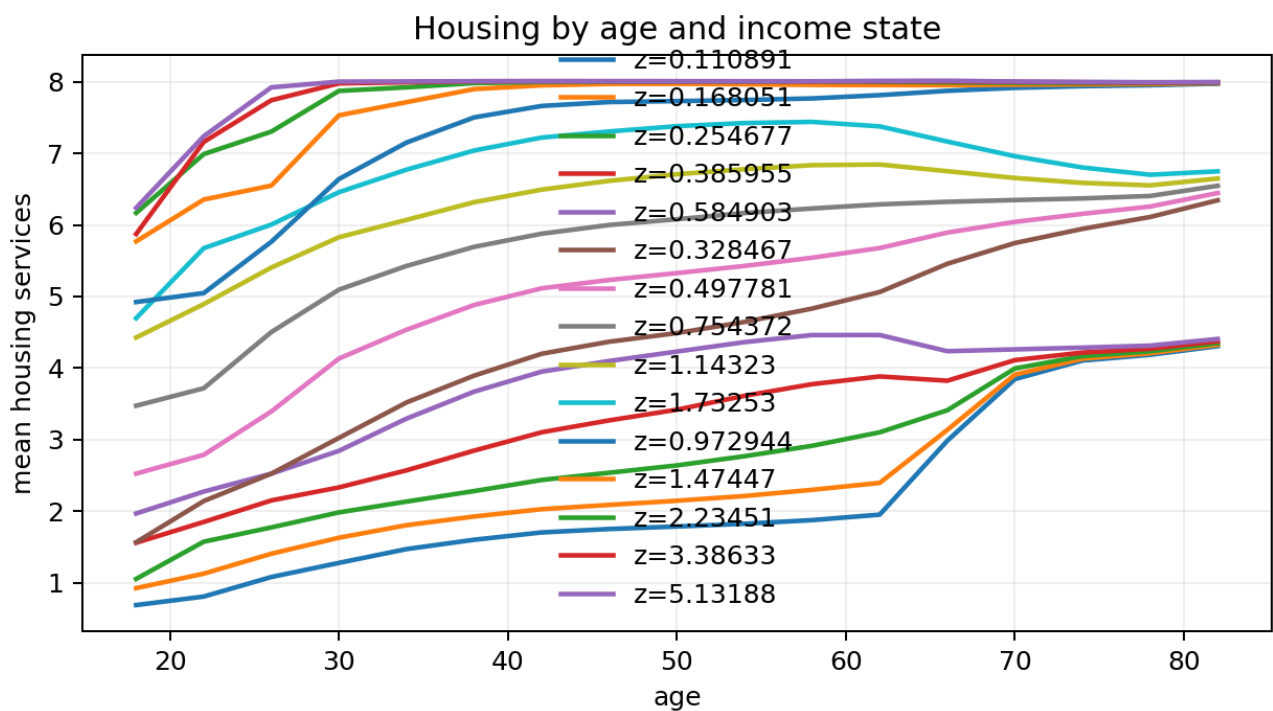
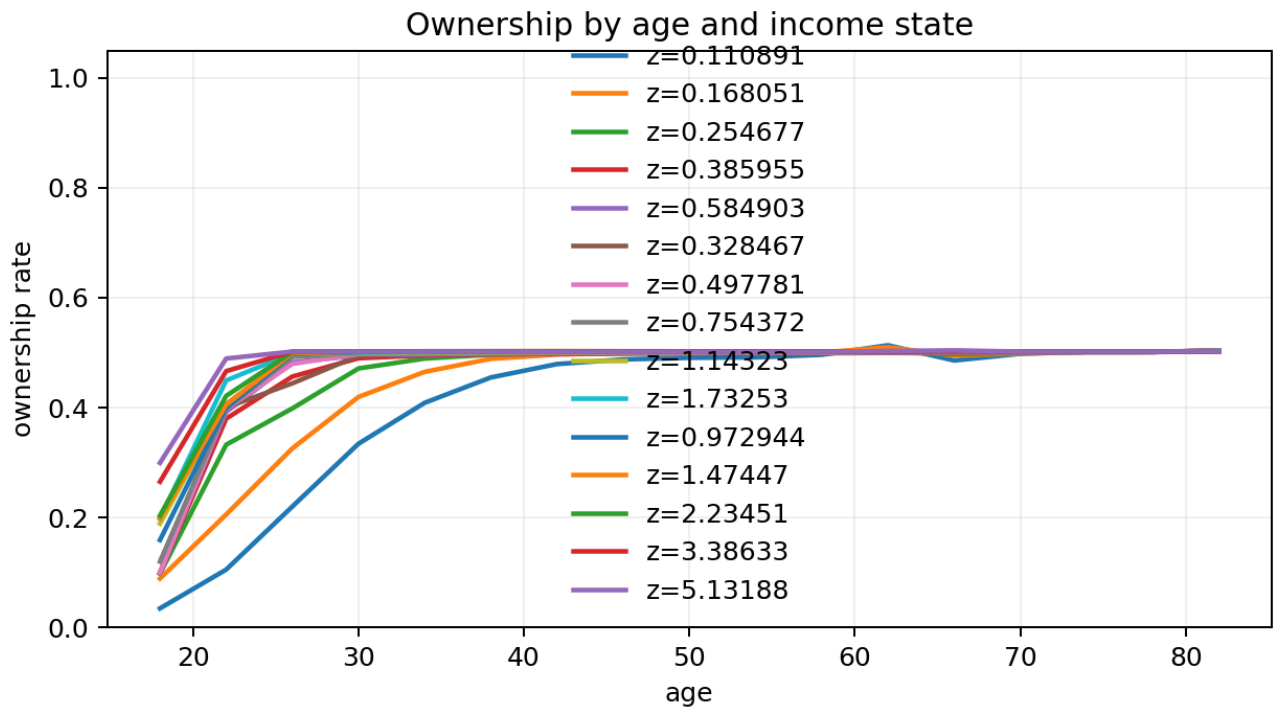
Free parameter	Starting value	Lower	Upper	Near bound?
Annual discount factor	0.995117	0.94	0.9995	False
Outer tenure scale κ	2	0.005	10	False
Dissimilarity λ	0.8	0.02	1	False
Owner housing-service premium χ	1.04347	0.1	5	False
Housing-supply normalization H_0	14.563	0.2	80	False
Bequest utility weight θ_0	0.528428	0	8	False
Bequest wealth shift θ_1	0.107249	0.02	16	True
Per-child room floor	0.28221	0.1	1.8	False
First-birth utility cost	4.55914	0	8	False
First-child room-floor jump	0.364931	0	2	False
2007-2023 child-value change	-0.328714	-1.5	0.2	False

Other object	Value	Restriction
Child value in 2007	0.28071	Normalized to old completed fertility = 2.1
Child value in 2023	-0.0480044	Derived from 2007 level plus fitted change
Supply elasticity	0.63	Externally fixed

Near-bound flags use the existing convention: within 2% of the full physical parameter range. Thus a flag for a log-transformed parameter need not imply a small distance in its search coordinate. The lower restriction $\lambda \geq 0.02$ is numerical; the upper restriction $\lambda \leq 1$ comes from the chosen nested-GEV law.

Standard policy diagnostics: starting-point weakness

The ownership gradient is weak at these starting scales. This is economically important despite passing probability, market-clearing and occupied-value checks. It does not show that the target is unreachable after recalibration.



Policy-loop check and limits of the current evidence

The verification stopped, so a complete policy comparison is unavailable. The completed history is reproducible; the policy path is not certified. No gate has been relaxed.

Policy	2023	2027
Baseline	Passed	Passed
Housing supply +20%	Passed	Occupied-value check failed
Dependent-child LTV 95%	Not run	Not run
Property tax doubled	Not run	Not run

The new failure. At age 18, a childless renter in income state 8 gains wealth from -1.79070 to -1.65116, but its computed value falls by 0.256796. The lower state contains 0.1313% of pre-choice household mass. The budget audit passes. This is separate from the corrected consumption-reporting floor.

Bounded numerical diagnosis. The local replay reproduces all thirteen policy/population arrays exactly. Exhaustive saving leaves 0 occupied value decreases. At the same prices and inherited population, births change by -0.0300701%, ownership by -0.000226077 percentage points and rooms by -0.0583628%. These are optimizer comparisons, not policy effects. The resulting market residual is 0.000616, above the 2e-4 clearing tolerance; prices must be solved again. Full historical and policy-path verification remains required.

Maintained closures. Markets clear separately at each date; households treat each current price as permanent. This is temporary equilibrium, not perfect foresight. After 2023 the national population is closed: no outside entry, retention one, the inherited four-slot birth queue and births converted into entrant households by 1/2.1. The property-tax experiment discards revenue and has no rebate or grant. These are finite-horizon paths, not stationary endpoints.

Verified population handoff. The original inherited population is saved separately from its price-specific feasibility adjustment. The fitted gate replays exactly; adjusted mass is 2.325e-15. The queue entering 2023 comes from the completed 2019 row, not the already advanced 2023 row.

Verified reporting repair. A legacy owner-consumption output floor reported 0.040 where one budget supported 0.015366. The experimental output now reports consumption implied by the optimizer. At the exact failed policy price, all fifteen value, choice and distribution arrays, market quantities and seventeen graphs are unchanged. Budget-violating mass falls from 1.94e-07 to 1.16e-12, below the unchanged 2e-10 gate.

Still outstanding. Full calibration; local identification and parameter trade-offs; exact final repetitions of a selected optimum; calibrated 2023-2063 policy paths; and author adoption of the nesting/commitment restrictions. No completed smoke settles those questions.

Evidence: `output/model/e5f_joint_nested_full_20260906a/README.md` indexes the revisions, immutable contracts, numerical checks, full tables and all seventeen unchanged standard graphs. Rebuild this PDF with `build_e5f_joint_nested_review.py` and the same collected history/policy directories.